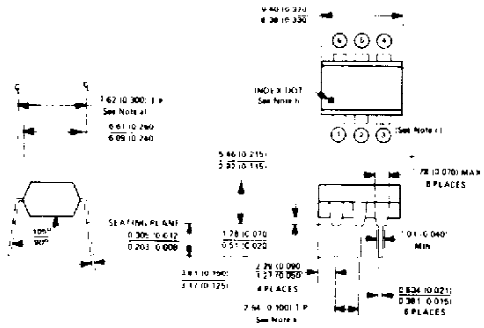
SOOS035 D2493 SEPTEMBER 1978 - REVISED MARCH 1983

COMPATIBLE WITH STANDARD TTL INTEGRATED CIRCUITS

- **Gallium Arsenide Diode Infrared Source Optically Coupled to a Silicon N-P-N Phototransistor**
- **High Direct-Current Transfer Ratio**
- **High-Voltage Electrical Isolation . . . 2.5-kV, 1.5-kV, or 0.5-kV Rating**
- **Plastic Dual-In-Line Package**
- **High-Speed Switching . . . t_r = 2 μs, t_f = 2 μs Typical**

mechanical data

The package consists of a gallium arsenide infrared-emitting diode and an n-p-n silicon phototransistor mounted on a 6-lead frame encapsulated within an electrically nonconductive plastic compound. The case will withstand soldering temperature with no deformation and device performance characteristics remain stable when operated in high-humidity conditions. Unit weight is approximately 0.52 grams.



NOTES:
a. Leads are within 0,127 (0.005) radius of true position (T.P.) with maximum material condition and unit installed.
b. Pin 1 identified by index dot.
c. Terminal connections:
1. Anode } Infrared-emitting diode
2. Cathode
3. No internal connection
4. Emitter } Phototransistor
5. Collector
6. Base

FALLS WITHIN JEDEC MO-001AM DIMENSIONS
ALL LINEAR DIMENSIONS ARE IN MILLIMETERS AND PARENTHETICALLY IN INCHES



absolute maximum ratings at 25°C free-air temperature (unless otherwise noted)

*Peak Input-to-Output Voltage: 4N25	±2.5 kV
4N26, 4N27	±1.5 kV
4N28	±0.5 kV
*Collector-Base Voltage	70 V
*Collector-Emitter Voltage (See Note 1)	30 V
*Emitter-Collector Voltage	7 V
Emitter-Base Voltage	7 V
*Input-Diode Reverse Voltage	3 V
*Input-Diode Continuous Forward Current at (or below) 25°C Free-Air Temperature (See Note 2)	80 mA
*Input-Diode Peak Forward Current (t_W = 300 μs, duty cycle = 2%)	3 A
*Continuous Power Dissipation at (or below) 25°C Free-Air Temperature:	
Infrared-Emitting Diode (See Note 3)	150 mW
Phototransistor (See Note 3)	150 mW
Total, Infrared-Emitting Diode plus Phototransistor (See Note 4)	250 mW
*Storage Temperature Range	−55°C to 150°C
*Lead Temperature 1,6 mm (1/16 inch) from Case for 10 Seconds	260°C

*JEDEC registered data. This data sheet contains all applicable JEDEC-registered data in effect at the time of publication.

NOTES: 1. This value applies when the base-emitter diode is open-circulated.
2. Derate linearly to 100°C free-air temperature at the rate of 1.33 mA/°C.
3. Derate linearly to 100°C free-air temperature at the rate of 2 mW/°C.
4. Derate linearly to 100°C free-air temperature at the rate of 3.33 mW/°C.

PRODUCTION DATA documents contain information current as of publication date. Products conform to specifications per the terms of Texas Instruments standard warranty. Production processing does not necessarily include testing of all parameters.

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electrical characteristics at 25°C free-air temperature (unless otherwise noted)

PARAMETER		TEST CONDITIONS	4N25, 4N26			4N27, 4N28			UNIT
			MIN	TYP	MAX	MIN	TYP	MAX	
*$V_{(BR)CBO}$	Collector-Base Breakdown Voltage	I_C = 100 μA, I_E = 0, I_F = 0	70			70			V
*$V_{(BR)CEO}$	Collector-Emitter Breakdown Voltage	I_C = 1 mA, I_B = 0, I_F = 0	30			30			V
*$V_{(BR)ECO}$	Emitter-Collector Breakdown Voltage	I_E = 100 μA, I_B = 0, I_F = 0	7			7			V
*I_R	Input Diode Static Reverse Current	V_R = 3 V			100			100	μA
*$I_{C(on)}$	On-State Collector Current (Phototransistor Operation)	V_{CE} = 10 V, I_B = 0, I_F = 10 mA	2	5		1	3		mA
$I_{C(on)}$	On-State Collector Current (Photodiode Operation)	V_{CB} = 10 V, I_E = 0, I_F = 10 mA		20			20		μA
*$I_{C(off)}$	Off-State Collector Current (Phototransistor Operation)	V_{CE} = 10 V, I_B = 0, I_F = 0		1	50		1	50	nA
*$I_{C(off)}$	Off-State Collector current (Photodiode Operation)	V_{CB} = 10 V, I_E = 0, I_F = 0		0.1	20		0.1	20	nA
*V_F	Input Diode Static Forward Voltage	I_F = 10 mA		1.25	1.5		1.25	1.5	V
*$V_{CE(sat)}$	Collector-Emitter Saturation Voltage	I_C = 2 mA, I_B = 0, I_F = 50 mA		0.25	0.5		0.25	0.5	V
r_{IO}	Input-to-Output Internal resistance	V_{in-out} = ±2.5 kV for 4N25, ±1.5 kV for 4N26, 4N27, ±0.5 kV for 4N28, See Note 5	10^{11}	10^{12}		10^{11}	10^{12}		Ω
C_{io}	Input-to-Output Capacitance	V_{in-out} = 0, f = 1 MHz, See Note 5		1			1		pF

*JEDEC registered data

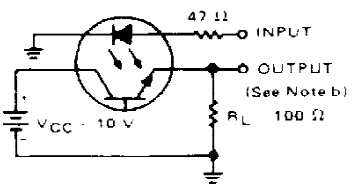
NOTE 5: These parameters are measured between both input diode leads shorted together and all the phototransistor leads shorted together

switching characteristics at 25°C free-air temperature

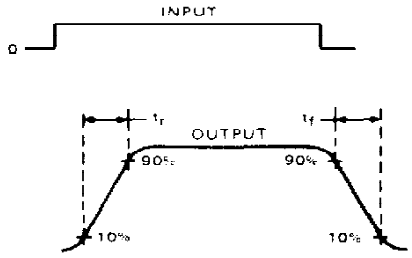
PARAMETER			TEST CONDITIONS	TYP	UNIT
t_r	Rise Time	Phototransistor Operation	V_{CC} = 10 V, I_B = 0, $I_{C(on)}$ = 2 mA, R_L = 100 Ω, See Test Circuit A of Figure 1	2	μs
t_f	Fall Time			2	
t_r	Rise Time	Photodiode Operation	V_{CC} = 10 V, I_E = 0, $I_{C(on)}$ = 20 μA, R_L = 1 kΩ, See Test Circuit B of Figure 1	1	μs
t_f	Fall Time			1	

PARAMETER MEASUREMENT INFORMATION

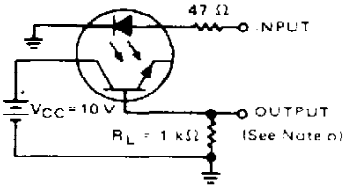
Adjust amplitude of input pulse for:
$I_{C(on)}$ = 2 mA (Test Circuit A) or
$I_{C(on)}$ = 20 μA (Test Circuit B)



TEST CIRCUIT A
PHOTOTRANSISTOR OPERATION



VOLTAGE WAVEFORMS



TEST CIRCUIT B
PHOTODIODE OPERATION

NOTES a. The input waveform is supplied by a generator with the following characteristics: Z_{out} = 50 Ω, t_r ≤ 15 ns, duty cycle ≈ 1%, t_w = 100 μs.
b. The output waveform is monitored on an oscilloscope with the following characteristics: t_r ≤ 12 ns, R_{in} ≥ 1 MΩ, C_{in} ≤ 20 pF.

FIGURE 1 – SWITCHING TIMES

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